To know the PFAS related chemicals better

PFOA

Perfluorooctanoic acid (PFOA) is a specific PFAS compound identified by the Chemical Abstract Services (CAS) Number 335-67-1.
Often referred to as "C8," PFOA is used as an industrial surfactant in chemical processes and as a material feedstock.
PFOA, along with Perfluorooctanesulfonic acid (PFOS), was one of the first PFAS compounds linked to human health effects and persistence in the environment and therefore regulated.

PFOS

PFOS is a specific PFAS compound identified by the CAS Number 1763-23-1.

It is a fluorosurfactant and was **previously the key ingredient in various fabric protection products.**
PFOS, along with PFOA, was one of the first PFAS compounds linked to human health effects and persistence in the environment and therefore regulated.

PFCs

PFCs Historically referred to Perfluorochemicals or Perfluorinated Compounds, this chemical class is now broadly referred to as "Perfluoroalkyl and Polyfluoroalkyl Substances" or "PFAS" instead.
The term "PFAS" is preferred since "PFCs" is also used in reference to "Perfluorocarbons", which are human made chemicals composed of carbon and fluorine only, and which are regulated due to being powerful greenhouse gases. Perfluorocarbons are separate and distinct from PFAS compounds, with different properties and hazard traits.

PFAS

Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) are synthetic chemicals defined as "fluorinated substances that contain at least one fully fluorinated methyl or methylene carbon atom (without any H/ Cl/Br/I atom attached to it), i.e., with a few noted exceptions, any chemical with at least a perfluorinated methyl group (–CF3) or a perfluorinated methylene group (–CF2–) is a PFAS."
This definition is provided by the Organization for Economic Co-operation and Development (OECD) which, along with the United States Environmental Protection

Agency (U.S EPA), defines several thousand substances as belonging to the group of PFAS.

New legislation in, e.g., California and New York, defines PFAS more broadly as "fluorinated organic chemicals containing at least one fully fluorinated carbon atom." **The challenging part is the current OECD, U.S EPA, and U.S. state definitions are not harmonized.**

Why Are PFAS Restricted? **AFIRM**

Many PFAS have been found to cause long-term health effects depending on the level and duration of exposure, including at very low levels.

PFAS are often referred to as "forever chemicals" because of **their resistance to degradation and resulting persistence in the environment, which is a function of the carbon-fluorine bonds they contain (one of the strongest single bonds in chemistry).** Due to their environmental persistence and wide range of hazard traits, authorities around the world are increasingly regulating the entire class of PFAS instead of specific subclasses or individual PFAS.

C8 refers to perfluorooctanoic acid (PFOA), which is a type of per- and polyfluoroalkyl substance (PFAS). PFOA is distinct from perfluorooctane sulfonate (PFOS), which is another type of PFAS.

Both PFOA and PFOS are banned.

C6 including PFHxA and PFHxS

According to the definitions in the latest amendment, PFHxS, its salts and PFHxS-related compounds means:

PFHxS including any of its branched isomers

Its salts

PFHxS-related compounds which are substances that degrade to PFHxS, including substances that contains the chemical moiety $C_6F_{13}S$- as one of its structural elements